



QUESTIONS AND ANSWER EXPLANATIONS: All of the private pilot knowledge test questions chosen by the FAA for release as well as additional questions selected by Gleim relating to the material in the previous outlines are provided on the following pages. These questions have been organized into the same subunits as the outlines. To the immediate right of each question are the correct answer and answer explanations. You should cover these answers and answer explanations while responding to the questions. Refer to the general discussion in the Introduction on how to take the FAA knowledge test.

Remember that the questions from the FAA knowledge test bank have been reordered by topic and organized into a meaningful sequence. Also, the first line of the answer explanation gives the citation of the authoritative source for the answer.

QUESTIONS

7.1 Causes of Weather

1. Every physical process of weather is accompanied by, or is the result of, a

- A. movement of air.
- B. pressure differential.
- C. heat exchange.

Answer (C) is correct. (FAA-H-8083-25B Chap 12)

DISCUSSION: Every physical process of weather is accompanied by, or is the result of, a heat exchange. A heat differential (difference between the temperatures of two air masses) causes a differential in pressure, which in turn causes movement of air. Heat exchanges occur constantly, e.g., melting, cooling, updrafts, downdrafts, wind, etc.

Answer (A) is incorrect. Movement of air is a result of heat exchange. **Answer (B) is incorrect.** Pressure differential is a result of heat exchange.

2. What causes variations in altimeter settings between weather reporting points?

- A. Unequal heating of the Earth's surface.
- B. Variation of terrain elevation.
- C. Coriolis force.

Answer (A) is correct. (FAA-H-8083-25B Chap 12)

DISCUSSION: Unequal heating of the Earth's surface causes differences in air pressure, which is reflected in differences in altimeter settings between weather reporting points.

Answer (B) is incorrect. Variations in altimeter settings between stations is a result of unequal heating of the Earth's surface, not variations of terrain elevations. **Answer (C) is incorrect.** Variations in altimeter settings between stations is a result of unequal heating of the Earth's surface, not the Coriolis force.

3. The wind at 5,000 feet AGL is southwesterly while the surface wind is southerly. This difference in direction is primarily due to

- A. stronger pressure gradient at higher altitudes.
- B. friction between the wind and the surface.
- C. stronger Coriolis force at the surface.

Answer (B) is correct. (AC 00-6B Chap 7)

DISCUSSION: Winds aloft at 5,000 ft. are largely affected by Coriolis force, which deflects wind to the right, in the Northern Hemisphere. But at the surface, the winds will be more southerly (they were southwesterly aloft) because Coriolis force has less effect at the surface where the wind speed is slower. The wind speed is slower at the surface due to the friction between the wind and the surface.

Answer (A) is incorrect. Pressure gradient is a force that causes wind, not the reason for wind direction differences.

Answer (C) is incorrect. The Coriolis force at the surface is weaker (not stronger) with slower wind speed.

7.2 Convective Currents

4. The development of thermals depends upon

- A. a counterclockwise circulation of air.
- B. temperature inversions.
- C. solar heating.

Answer (C) is correct. (AC 00-6B Chap 4)

DISCUSSION: Thermals are updrafts in small-scale convective currents. Convective currents are caused by uneven heating of the Earth's surface. Solar heating is the means of heating the Earth's surface.

Answer (A) is incorrect. A counterclockwise circulation describes an area of low pressure in the Northern Hemisphere.

Answer (B) is incorrect. A temperature inversion is an increase in temperature with height, which hinders the development of thermals.

5. Convective circulation patterns associated with sea breezes are caused by

- A. warm, dense air moving inland from over the water.
- B. water absorbing and radiating heat faster than the land.
- C. cool, dense air moving inland from over the water.

Answer (C) is correct. (AC 00-6B Chap 9)

DISCUSSION: Sea breezes are caused by cool and more dense air moving inland off the water. Once over the warmer land, the air heats up and rises. Thus the cooler, more dense air from the sea forces the warmer air up. Currents push the hot air over the water where it cools and descends, starting the cycle over again. This process is caused by land heating faster than water.

Answer (A) is incorrect. The air over the water is cooler (not warmer). **Answer (B) is incorrect.** Water absorbs and radiates heat slower (not faster) than land.

7.3 Fronts

6. The boundary between two different air masses is referred to as a

- A. frontolysis.
- B. frontogenesis.
- C. front.

Answer (C) is correct. (AC 00-6B Chap 10)

DISCUSSION: A front is a surface, interface, or transition zone of discontinuity between two adjacent air masses of different densities. It is the boundary between two different air masses.

Answer (A) is incorrect. Frontolysis is the dissipation of a front. **Answer (B) is incorrect.** Frontogenesis is the initial formation of a front or frontal zone.

7. One weather phenomenon which will always occur when flying across a front is a change in the

- A. wind direction.
- B. type of precipitation.
- C. stability of the air mass.

Answer (A) is correct. (AC 00-6B Chap 10)

DISCUSSION: The definition of a front is the zone of transition between two air masses of different air pressure or density, e.g., the area separating high and low pressure systems. Due to the difference in changes in pressure systems, there will be a change in wind.

Answer (B) is incorrect. Frequently, precipitation will exist or not exist for both sides of the front: rain showers before and after or no precipitation before and after a dry front.

Answer (C) is incorrect. Fronts separate air masses with different pressures, not stabilities; e.g., both air masses could be either stable or unstable.

8. One of the most easily recognized discontinuities across a front is

- A. a change in temperature.
- B. an increase in cloud coverage.
- C. an increase in relative humidity.

Answer (A) is correct. (AC 00-6B Chap 10)

DISCUSSION: Of the many changes that take place across a front, the most easily recognized is the change in temperature. When flying through a front, you will notice a significant change in temperature, especially at low altitudes.

Answer (B) is incorrect. Although cloud formations may indicate a frontal system, they may not be present or easily recognized across the front. **Answer (C) is incorrect.** Precipitation is not always associated with a front.

7.4 Thunderstorms

9. If there is thunderstorm activity in the vicinity of an airport at which you plan to land, which hazardous atmospheric phenomenon might be expected on the landing approach?

- A. Precipitation static.
- B. Wind-shear turbulence.
- C. Steady rain.

10. A nonfrontal, narrow band of active thunderstorms that often develop ahead of a cold front is known as a

- A. prefrontal system.
- B. squall line.
- C. dry line.

11. What conditions are necessary for the formation of thunderstorms?

- A. High humidity, lifting force, and unstable conditions.
- B. High humidity, high temperature, and cumulus clouds.
- C. Lifting force, moist air, and extensive cloud cover.

12. During the life cycle of a thunderstorm, which stage is characterized predominately by downdrafts?

- A. Cumulus.
- B. Dissipating.
- C. Mature.

13. Thunderstorms reach their greatest intensity during the

- A. mature stage.
- B. downdraft stage.
- C. cumulus stage.

14. What feature is normally associated with the cumulus stage of a thunderstorm?

- A. Roll cloud.
- B. Continuous updraft.
- C. Frequent lightning.

Answer (B) is correct. (AC 00-6B Chap 19)

DISCUSSION: The most hazardous atmospheric phenomenon near thunderstorms is wind shear turbulence.

Answer (A) is incorrect. Precipitation static is a steady, high level of noise in radio receivers, which is caused by intense corona discharges from sharp metallic points and edges of flying aircraft. This discharge may be seen at night and is also called St. Elmo's fire. **Answer (C) is incorrect.** Thunderstorms are usually associated with unstable air, which would produce rain showers (not steady rain).

Answer (B) is correct. (AC 00-6B Chap 19)

DISCUSSION: A nonfrontal, narrow band of active thunderstorms that often develops ahead of a cold front is known as a squall line.

Answer (A) is incorrect. A prefrontal system is a term that has no meaning. **Answer (C) is incorrect.** A dry line is a front that seldom has any significant air mass contrast except for moisture.

Answer (A) is correct. (AC 00-6B Chap 19)

DISCUSSION: Thunderstorms form when there is sufficient water vapor, an unstable lapse rate, and an initial upward boost (lifting) to start the storm process.

Answer (B) is incorrect. A high temperature is not required for the formation of thunderstorms. **Answer (C) is incorrect.** Extensive cloud cover is not necessary for the formation of thunderstorms.

Answer (B) is correct. (AC 00-6B Chap 19)

DISCUSSION: Thunderstorms have three life cycles: cumulus, mature, and dissipating. It is in the dissipating stage that the storm is characterized by downdrafts as the storm rains itself out.

Answer (A) is incorrect. Cumulus is the building stage when there are updrafts. **Answer (C) is incorrect.** The mature stage is when there are both updrafts and downdrafts, which create dangerous wind shears.

Answer (A) is correct. (AC 00-6B Chap 19)

DISCUSSION: Thunderstorms reach their greatest intensity during the mature stage, where updrafts and downdrafts cause a high level of wind shear.

Answer (B) is incorrect. The downdraft stage is known as the dissipating stage, which is when the thunderstorm rains itself out. **Answer (C) is incorrect.** The cumulus stage is characterized by continuous updrafts and is not the most intense stage of a thunderstorm.

Answer (B) is correct. (AC 00-6B Chap 19)

DISCUSSION: The cumulus stage of a thunderstorm has continuous updrafts that build the storm. The water droplets are carried up until they become too heavy. Once they begin falling and creating downdrafts, the storm changes from the cumulus to the mature stage.

Answer (A) is incorrect. The roll cloud is the cloud on the ground, which is formed by the downrushing cold air pushing out from underneath the bottom of the thunderstorm.

Answer (C) is incorrect. Frequent lightning is associated with the mature stage, where there is a considerable amount of wind shear and static electricity.

15. Which weather phenomenon signals the beginning of the mature stage of a thunderstorm?

- A. The appearance of an anvil top.
- B. Precipitation beginning to fall.
- C. Maximum growth rate of the clouds.

16. Thunderstorms which generally produce the most intense hazard to aircraft are

- A. squall line thunderstorms.
- B. steady-state thunderstorms.
- C. warm front thunderstorms.

17. Which weather phenomenon is always associated with a thunderstorm?

- A. Lightning.
- B. Heavy rain.
- C. Hail.

18. The mature stage of a thunderstorm begins with

- A. formation of the anvil top.
- B. the start of precipitation.
- C. continuous downdrafts.

19. The destination airport has one runway, 8-26, and the wind is calm. The normal approach in calm wind is a left-hand pattern to runway 8. There is no other traffic at the airport. A thunderstorm about 6 miles west is beginning its mature stage, and rain is starting to reach the ground. The pilot decides to

- A. depart expecting the thunderstorm to dissipate prior to arrival, then land on runway 8.
- B. delay departure until the thunderstorm has dissipated.
- C. fly an approach to runway 26 since any unexpected wind due to the storm will be westerly.

Answer (B) is correct. (AC 00-6B Chap 19)

DISCUSSION: The mature stage of a thunderstorm begins when rain begins falling. This means that the downdrafts are occurring sufficiently to carry water all the way through the thunderstorm.

Answer (A) is incorrect. The appearance of an anvil top normally occurs during the dissipating stage when the upper winds blow the top of the cloud downwind. **Answer (C) is incorrect.** The maximum growth rate of clouds is later in the mature stage and does not necessarily mark the start of the mature stage.

Answer (A) is correct. (AC 00-6B Chap 19)

DISCUSSION: A squall line is a nonfrontal, narrow band of active thunderstorms. It often contains severe, steady-state thunderstorms and presents the single most intense weather hazard to aircraft.

Answer (B) is incorrect. Steady-state thunderstorms are normally associated with weather systems and often form into squall lines. **Answer (C) is incorrect.** Squall line (not warm front) thunderstorms generally produce the most intense hazard to aircraft.

Answer (A) is correct. (AC 00-6B Chap 19)

DISCUSSION: A thunderstorm, by definition, has lightning, because lightning causes the thunder.

Answer (B) is incorrect. While heavy rain showers usually occur, hail may occur instead. **Answer (C) is incorrect.** Hail is produced only when the lifting action extends above the freezing level and the supercooled water begins to freeze.

Answer (B) is correct. (AC 00-6B Chap 19)

DISCUSSION: The mature stage of a thunderstorm begins when rain begins falling. This means that the downdrafts are occurring sufficiently to carry water all the way through the thunderstorm.

Answer (A) is incorrect. The formation of an anvil top normally occurs during the dissipating stage when the upper winds blow the top of the cloud downwind. **Answer (C) is incorrect.** Continuous downdrafts are present during the dissipating stage of a thunderstorm, not during the mature stage.

Answer (B) is correct. (FAA-H-8083-25B Chap 12)

DISCUSSION: During the mature stage of a thunderstorm, precipitation descends through the cloud and drags the adjacent air downward, creating a strong downdraft. The downdraft spreads out along the surface, well in advance of the parent thunderstorm cell, as a mass of cool, gusty air. Adverse winds always are found within thunderstorms and often many miles from the precipitation area. Crosswinds, gusts, and variable winds (i.e., sudden wind shifts) can lead to a crash during takeoffs, approaches, and landings. Therefore, the best decision would be to delay departure from the departure airport until the thunderstorm has dissipated at the arrival airport.

Answer (A) is incorrect. Adverse winds always are found within thunderstorms and often many miles from the precipitation area. Crosswinds, gusts, and variable winds (i.e., sudden wind shifts) can lead to a crash during takeoffs, approaches, and landings. The best decision would be to delay departure from the departure airport until the thunderstorm has dissipated at the arrival airport. **Answer (C) is incorrect.** Adverse winds always are found within thunderstorms and often many miles from the precipitation area. Crosswinds, gusts, and variable winds (i.e., sudden wind shifts) can lead to a crash during takeoffs, approaches, and landings. The best decision would be to deviate to an alternate airport, not fly an approach into the airport.

20. You are avoiding a thunderstorm that is in your flightpath. You are over 20 miles from the cell; however, you are under the anvil of the cell. Is this a hazard?

- A. No, you are at a safe distance from the cell.
- B. Yes, hail can be discharged from the anvil.
- C. Yes, this is still in the area of dissipation.

7.5 Icing

21. One in-flight condition necessary for structural icing to form is

- A. small temperature/dewpoint spread.
- B. stratiform clouds.
- C. visible moisture.

22. How will frost on the wings of an airplane affect takeoff performance?

- A. Frost will disrupt the smooth flow of air over the wing, adversely affecting its lifting capability.
- B. Frost will change the camber of the wing, increasing its lifting capability.
- C. Frost will cause the airplane to become airborne with a higher angle of attack, decreasing the stall speed.

23. Which conditions result in the formation of frost?

- A. The temperature of the collecting surface is at or below freezing when small droplets of moisture fall on the surface.
- B. The temperature of the collecting surface is at or below the dewpoint of the adjacent air and the dewpoint is below freezing.
- C. The temperature of the surrounding air is at or below freezing when small drops of moisture fall on the collecting surface.

24. How does frost affect the lifting surfaces of an airplane on takeoff?

- A. Frost may prevent the airplane from becoming airborne at normal takeoff speed.
- B. Frost will change the camber of the wing, increasing lift during takeoff.
- C. Frost may cause the airplane to become airborne with a lower angle of attack at a lower indicated airspeed.

Answer (B) is correct. (AC 00-24C, AIM Para 7-1-26)

DISCUSSION: Pilots should anticipate possible hail with any thunderstorm but especially beneath the anvil of a large cumulonimbus.

Answer (A) is incorrect. Even if your flightpath is over 20 mi. from the cell, there is still potential to encounter hazards such as extreme clear air turbulence and hail when under the anvil. **Answer (C) is incorrect.** When under the anvil, the hazards you are most likely to encounter are clear air turbulence and hail.

Answer (C) is correct. (AC 00-6B Chap 18)

DISCUSSION: Two conditions are necessary for structural icing while in flight. First, the airplane must be flying through visible moisture, such as rain or cloud droplets. Second, the temperature at the point where the moisture strikes the airplane must be freezing or below.

Answer (A) is incorrect. The temperature dew point spread is not a factor in icing as it is in the formation of fog or clouds. **Answer (B) is incorrect.** No special cloud formation is necessary for icing as long as visible moisture is present.

Answer (A) is correct. (FAA-H-8083-25B Chap 12)

DISCUSSION: Frost does not change the basic aerodynamic shape of the wing, but the roughness of its surface spoils the smooth flow of air, thus causing an increase in drag and an early airflow separation over the wing, resulting in a loss of lift.

Answer (B) is incorrect. Frost will decrease (not increase) lift during takeoff and has no effect on the wing camber.

Answer (C) is incorrect. A layer of frost on an airplane will increase drag, which increases (not decreases) the stall speed.

Answer (B) is correct. (AC 00-6B Chap 3)

DISCUSSION: Frost forms when both the collecting surface is below the dew point of the adjacent air and the dew point is below freezing. Frost is the deposition of water vapor to ice crystals.

Answer (A) is incorrect. If small droplets of water fall on the collecting surface, which is at or below freezing, ice (not frost) will form. **Answer (C) is incorrect.** If small droplets of water fall while the surrounding air is at or below freezing, ice (not frost) will form.

Answer (A) is correct. (FAA-H-8083-25B Chap 12)

DISCUSSION: Frost that is not removed from the surface of an airplane prior to takeoff may make it difficult to get the airplane airborne at normal takeoff speed. The frost disrupts the airflow over the wing, which increases drag.

Answer (B) is incorrect. The smoothness of the wing, not its curvature, is affected and lift is decreased (not increased).

Answer (C) is incorrect. Ground effect (not frost) may cause an airplane to become airborne with a lower angle of attack at a lower indicated airspeed.

25. In which environment is aircraft structural ice most likely to have the highest accumulation rate?

- A. Cumulus clouds with below freezing temperatures.
- B. Freezing drizzle.
- C. Freezing rain.

26. The presence of ice pellets at the surface is evidence that there

- A. are thunderstorms in the area.
- B. has been cold frontal passage.
- C. is a temperature inversion with freezing rain at a higher altitude.

27. Why is frost considered hazardous to flight?

- A. Frost changes the basic aerodynamic shape of the airfoils, thereby increasing lift.
- B. Frost slows the airflow over the airfoils, thereby increasing control effectiveness.
- C. Frost spoils the smooth flow of air over the wings, thereby decreasing lifting capability.

7.6 Mountain Wave

28. An almond or lens-shaped cloud which appears stationary, but which may contain winds of 50 knots or more, is referred to as

- A. an inactive frontal cloud.
- B. a funnel cloud.
- C. a lenticular cloud.

29. Crests of standing mountain waves may be marked by stationary, lens-shaped clouds known as

- A. mammatocumulus clouds.
- B. standing lenticular clouds.
- C. roll clouds.

30. Possible mountain wave turbulence could be anticipated when winds of 40 knots or greater blow

- A. across a mountain ridge, and the air is stable.
- B. down a mountain valley, and the air is unstable.
- C. parallel to a mountain peak, and the air is stable.

Answer (C) is correct. (AC 00-6B Chap 18)

DISCUSSION: Freezing rain usually causes the highest accumulation rate of structural icing because of the nature of the supercooled water striking the airplane.

Answer (A) is incorrect. While icing potential is great in cumulus clouds with below freezing temperatures, the highest accumulation rate is in an area with large, supercooled water drops (i.e., freezing rain). **Answer (B) is incorrect.** Freezing drizzle will not build up ice as quickly as freezing rain.

Answer (C) is correct. (AC 00-6B Chap 14, 18)

DISCUSSION: Rain falling through colder air may freeze during its descent, falling as ice pellets. Ice pellets always indicate freezing rain at a higher altitude.

Answer (A) is incorrect. Ice pellets form when rain freezes during its descent, which may or may not be as a result of a thunderstorm. **Answer (B) is incorrect.** Ice pellets only indicate that rain is freezing at a higher altitude, not that a cold front has passed through an area.

Answer (C) is correct. (FAA-H-8083-25B Chap 12)

DISCUSSION: Frost does not change the basic aerodynamic shape of the wing, but the roughness of its surface spoils the smooth flow of air, thus causing an increase in drag and an early airflow separation over the wing, resulting in a loss of lift.

Answer (A) is incorrect. Frost is thin and does not change the basic aerodynamic shape of the airfoil. Frost causes a decrease in lift. **Answer (B) is incorrect.** The smooth flow of air over the airfoil is affected, not control effectiveness.

Answer (C) is correct. (AC 00-6B Chap 17)

DISCUSSION: Lenticular clouds are lens-shaped and indicate the crests of standing mountain waves. They form in the updraft and dissipate in the downdraft, so they do not move as the wind blows through them. Lenticular clouds may contain winds of 50 kt. or more and are extremely dangerous.

Answer (A) is incorrect. Frontal clouds usually do not contain winds of 50 kt. or more, and if they do, they do not appear stationary. **Answer (B) is incorrect.** A funnel cloud is not stationary.

Answer (B) is correct. (AC 00-6B Chap 17)

DISCUSSION: Lens-shaped clouds, which indicate crests of standing mountain waves, are called standing lenticular clouds. They form in the updraft and dissipate in the downdraft so that they do not move as the wind blows through them.

Answer (A) is incorrect. Cumulonimbus mamma clouds (also mammatocumulus) are cumulonimbus clouds with pods or circular domes on the bottom that indicate severe turbulence.

Answer (C) is incorrect. Roll clouds are low-level, turbulent areas in the shear zone between the plow wind surrounding the outrushing air from a thunderstorm and the surrounding air.

Answer (A) is correct. (AC 00-6B Chap 17)

DISCUSSION: Always anticipate possible mountain wave turbulence when the air is stable and winds of 40 kt. or greater blow across a mountain or ridge.

Answer (B) is incorrect. The wind must blow across the mountain or ridge before it flows down a valley. The air must also be stable. If the air is unstable and produces convective turbulence, it will rise and disrupt the "wave." **Answer (C) is incorrect.** Any time the winds are 40 kt. or more and blowing across (not parallel) to the mountains, you should anticipate mountain wave turbulence.

7.7 Wind Shear Avoidance

31. Where does wind shear occur?

- A. Only at higher altitudes.
- B. Only at lower altitudes.
- C. At all altitudes, in all directions.

32. A pilot can expect a wind-shear zone in a temperature inversion whenever the wind speed at 2,000 to 4,000 feet above the surface is at least

- A. 10 knots.
- B. 15 knots.
- C. 25 knots.

33. When may hazardous wind shear be expected?

- A. When stable air crosses a mountain barrier where it tends to flow in layers forming lenticular clouds.
- B. In areas of low-level temperature inversion, frontal zones, and clear air turbulence.
- C. Following frontal passage when stratocumulus clouds form indicating mechanical mixing.

7.8 Temperature/Dew Point and Fog

34. If the temperature/dewpoint spread is small and decreasing, and the temperature is 62°F, what type weather is most likely to develop?

- A. Freezing precipitation.
- B. Thunderstorms.
- C. Fog or low clouds.

35. What is meant by the term "dewpoint"?

- A. The temperature at which condensation and evaporation are equal.
- B. The temperature at which dew will always form.
- C. The temperature to which air must be cooled to become saturated.

Answer (C) is correct. (AC 00-6B Chap 17)**DISCUSSION:** Wind shear is the eddies in between two wind currents of differing velocities, direction, or both. Wind shear may be associated with either a wind shift or a wind speed gradient at any level in the atmosphere.**Answer (A) is incorrect.** A wind shear may occur at any (not only higher) altitudes. **Answer (B) is incorrect.** A wind shear may occur at any (not only lower) altitudes.**Answer (C) is correct.** (AC 00-6B Chap 17)**DISCUSSION:** When taking off or landing in calm wind under clear skies within a few hours before or after sunset, prepare for a temperature inversion near the ground. You can be relatively certain of a shear zone in the inversion if you know the wind is 25 kt. or more at 2,000 to 4,000 feet. Allow a margin of airspeed above normal climb or approach speed to alleviate the danger of stall in the event of turbulence or sudden change in wind velocity.**Answer (A) is incorrect.** A wind shear zone can be expected in a temperature inversion at 2,000 to 4,000 ft. AGL if the wind speed is at least 25 kt., not 10 kt. **Answer (B) is incorrect.** A wind shear zone can be expected in a temperature inversion at 2,000 to 4,000 ft. AGL if the wind speed is at least 25 kt., not 15 kt.**Answer (B) is correct.** (AC 00-6B Chap 17)**DISCUSSION:** Wind shear is the abrupt rate of change of wind velocity (direction and/or speed) per unit of distance and is normally expressed as vertical or horizontal wind shear. Hazardous wind shear may be expected in areas of low-level temperature inversion, frontal zones, and clear air turbulence.**Answer (A) is incorrect.** A mountain wave forms when stable air crosses a mountain barrier where it tends to flow in layers forming lenticular clouds. Turbulence, not wind shear, is expected in this area. **Answer (C) is incorrect.** Mechanical turbulence (not wind shear) may be expected following frontal passage when clouds form, indicating mechanical mixing.**Answer (C) is correct.** (AC 00-6B Chap 16)**DISCUSSION:** The difference between the air temperature and dew point is the temperature/dew point spread. As the temperature/dew point spread decreases, fog or low clouds tend to develop.**Answer (A) is incorrect.** There cannot be freezing precipitation if the temperature is 62°F. **Answer (B) is incorrect.** Thunderstorms have to do with unstable lapse rates, not temperature/dew point spreads.**Answer (C) is correct.** (AC 00-6B Chap 3)**DISCUSSION:** Dew point is the temperature to which air must be cooled to become saturated or have 100% humidity.**Answer (A) is incorrect.** Evaporation is the change from water to water vapor and is not directly related to the dew point. **Answer (B) is incorrect.** Dew forms only when heat radiates from an object whose temperature lowers below the dew point of the adjacent air.

36. The amount of water vapor which air can hold depends on the

- A. dewpoint.
- B. air temperature.
- C. stability of the air.

37. What are the processes by which moisture is added to unsaturated air?

- A. Evaporation and sublimation.
- B. Heating and condensation.
- C. Supersaturation and evaporation.

38. Clouds, fog, or dew will always form when

- A. water vapor condenses.
- B. water vapor is present.
- C. relative humidity reaches 100 percent.

39. Low-level turbulence can occur and icing can become hazardous in which type of fog?

- A. Rain-induced fog.
- B. Upslope fog.
- C. Steam fog.

40. In which situation is advection fog most likely to form?

- A. A warm, moist air mass on the windward side of mountains.
- B. An air mass moving inland from the coast in winter.
- C. A light breeze blowing colder air out to sea.

41. What situation is most conducive to the formation of radiation fog?

- A. Warm, moist air over low, flatland areas on clear, calm nights.
- B. Moist, tropical air moving over cold, offshore water.
- C. The movement of cold air over much warmer water.

Answer (B) is correct. (AC 00-6B Chap 3)**DISCUSSION:** Air temperature largely determines how much water vapor can be held by the air. Warm air can hold more water vapor than cool air.**Answer (A) is incorrect.** Dew point is the temperature at which air must be cooled to become saturated by the water vapor already present in the air. **Answer (C) is incorrect.** Air stability is the state of the atmosphere at which vertical distribution of temperature is such that air particles will resist displacement from their initial level.**Answer (A) is correct.** (AC 00-6B Chap 3)**DISCUSSION:** Evaporation is the process of converting a liquid to water vapor, and sublimation is the process of converting ice to water vapor.**Answer (B) is incorrect.** Heating alone does not add moisture. Condensation is the change of water vapor to liquid water. **Answer (C) is incorrect.** Supersaturation is a nonsense term in this context.**Answer (A) is correct.** (AC 00-6B Chap 3)**DISCUSSION:** As water vapor condenses, it becomes visible as clouds, fog, or dew.**Answer (B) is incorrect.** Water vapor is usually always present but does not form clouds, fog, or dew without condensation. **Answer (C) is incorrect.** Even at 100% humidity, water vapor may not condense, e.g., sufficient condensation nuclei may not be present.**Answer (C) is correct.** (AC 00-6B Chap 16)**DISCUSSION:** Steam fog forms in winter when cold, dry air passes from land areas over comparatively warm ocean waters and is composed entirely of water droplets that often freeze quickly. Low-level turbulence can occur, and icing can become hazardous.**Answer (A) is incorrect.** Precipitation- (rain-) induced fog is formed when relatively warm rain or drizzle falls through cool air and evaporation from the precipitation saturates the cool air and forms fog. While the hazards of turbulence and icing may occur in the proximity of rain-induced fog, these hazards occur as a result of the steam fog formation process. **Answer (B) is incorrect.** Upslope fog forms when moist, stable air is cooled as it moves up sloping terrain.**Answer (B) is correct.** (AC 00-6B Chap 16)**DISCUSSION:** Advection fog forms when moist air moves over colder ground or water. It is most common in coastal areas.**Answer (A) is incorrect.** A warm, moist air mass on the windward side of mountains produces rain or upslope fog as it blows upward and cools. **Answer (C) is incorrect.** A light breeze blowing colder air out to sea causes steam fog.**Answer (A) is correct.** (AC 00-6B Chap 16)**DISCUSSION:** Radiation fog is shallow fog of which ground fog is one form. It occurs under conditions of clear skies, little or no wind, and a small temperature/dew point spread. The fog forms almost exclusively at night or near dawn as a result of terrestrial radiation cooling the ground and the ground cooling the air on contact with it.**Answer (B) is incorrect.** Moist, tropical air moving over cold, offshore water causes advection fog, not radiation fog. **Answer (C) is incorrect.** Movement of cold, dry air over much warmer water results in steam fog.

42. What types of fog depend upon wind in order to exist?

- A. Radiation fog and ice fog.
- B. Steam fog and ground fog.
- C. Advection fog and upslope fog.

7.9 Clouds

43. Clouds are divided into four families according to their

- A. outward shape.
- B. height range.
- C. composition.

44. The suffix "nimbus," used in naming clouds, means

- A. a cloud with extensive vertical development.
- B. a rain cloud.
- C. a middle cloud containing ice pellets.

45. The conditions necessary for the formation of cumulonimbus clouds are a lifting action and

- A. unstable air containing an excess of condensation nuclei.
- B. unstable, moist air.
- C. either stable or unstable air.

46. What clouds have the greatest turbulence?

- A. Towering cumulus.
- B. Cumulonimbus.
- C. Nimbostratus.

47. At approximately what altitude above the surface would the pilot expect the base of cumuliiform clouds if the surface air temperature is 82°F and the dewpoint is 38°F?

- A. 9,000 feet AGL.
- B. 10,000 feet AGL.
- C. 11,000 feet AGL.

Answer (C) is correct. (AC 00-6B Chap 16)

DISCUSSION: Advection fog forms when moist air moves over colder ground or water. It is most common in coastal areas. Upslope fog forms when wind blows moist air upward over rising terrain and the air cools below its dew point. Both advection fog and upslope fog require wind to move air masses.

Answer (A) is incorrect. No wind is required for the formation of either radiation (ground) or ice fog. **Answer (B) is incorrect.** No wind is required for the formation of ground (radiation) fog.

Answer (B) is correct. (FAA-H-8083-25B Chap 12)

DISCUSSION: The four families of clouds are high clouds, middle clouds, low clouds, and clouds with extensive vertical development. Thus, they are based upon their height range.

Answer (A) is incorrect. Clouds are divided by their height range, not outward shape. **Answer (C) is incorrect.** Clouds are divided by their height range, not their composition.

Answer (B) is correct. (AC 00-6B Chap 13)

DISCUSSION: The suffix "nimbus" or the prefix "nimbo" means a rain cloud.

Answer (A) is incorrect. Clouds with extensive vertical development are called either towering cumulus or cumulonimbus. **Answer (C) is incorrect.** A middle cloud has the prefix "alto."

Answer (B) is correct. (AC 00-6B Chap 19)

DISCUSSION: Unstable, moist air, in addition to a lifting action, i.e., convective activity, is needed to form cumulonimbus clouds.

Answer (A) is incorrect. There must be moisture available to produce the clouds and rain; e.g., in a hot, dry dust storm, there would be no thunderstorm. **Answer (C) is incorrect.** The air must be unstable or there will be no lifting action.

Answer (B) is correct. (AC 00-6B Chap 13)

DISCUSSION: The greatest turbulence occurs in cumulonimbus clouds, which are thunderstorm clouds.

Answer (A) is incorrect. Towering cumulus clouds are an earlier stage of cumulonimbus clouds. **Answer (C) is incorrect.** Nimbostratus is a gray or dark, massive cloud layer diffused by continuous rain or ice pellets. It is a middle cloud with very little turbulence but may pose serious icing problems.

Answer (B) is correct. (FAA-H-8083-25B Chap 12)

DISCUSSION: The height of cumuliiform cloud bases can be estimated using surface temperature/dew point spread. Unsaturated air in a convective current cools at about 5.4°F/1,000 feet, and dew point decreases about 1°F/1,000 feet. In a convective current, temperature and dew point converge at about 4.4°F/1,000 feet. Thus, if the temperature/dew point spread is 44°F (82°F - 38°F), divide 44 by 4.4 to obtain 10,000 feet AGL.

Answer (A) is incorrect. The approximate height of the base of cumuliiform clouds is 9,000 feet AGL if the temperature/dew point spread is 40°F. **Answer (C) is incorrect.** The approximate height of the base of cumuliiform clouds is 11,000 feet AGL if the temperature/dew point spread is 48°F.

48. What cloud types would indicate convective turbulence?

- A. Cirrus clouds.
- B. Nimbostratus clouds.
- C. Towering cumulus clouds.

49. What is the approximate base of the cumulus clouds if the surface air temperature at 1,000 feet MSL is 70°F and the dewpoint is 48°F?

- A. 4,000 feet MSL.
- B. 5,000 feet MSL.
- C. 6,000 feet MSL.

50. Cumulus clouds often indicate

- A. possible turbulence.
- B. a temperature inversion.
- C. a dry adiabatic lapse rate.

7.10 Stability of Air Masses

51. What is a characteristic of stable air?

- A. Stratiform clouds.
- B. Unlimited visibility.
- C. Cumulus clouds.

52. When warm, moist, stable air flows upslope, it

- A. produces stratus type clouds.
- B. causes showers and thunderstorms.
- C. develops convective turbulence.

53. If an unstable air mass is forced upward, what type clouds can be expected?

- A. Stratus clouds with little vertical development.
- B. Stratus clouds with considerable associated turbulence.
- C. Clouds with considerable vertical development and associated turbulence.

Answer (C) is correct. (AC 00-6B Chap 13)

DISCUSSION: Towering cumulus clouds are an early stage of cumulonimbus clouds, or thunderstorms, that are based on convective turbulence, i.e., an unstable lapse rate.

Answer (A) is incorrect. Cirrus clouds are high, thin, featherlike ice crystal clouds in patches and narrow bands that are not based on any convective activity. **Answer (B) is incorrect.** Nimbostratus are gray or dark, massive clouds diffused by continuous rain or ice pellets with very little turbulence.

Answer (C) is correct. (FAA-H-8083-25B Chap 12)

DISCUSSION: The height of cumuliiform cloud bases can be estimated using surface temperature/dew point spread. Unsaturated air in a convective current cools at about 5.4°F/1,000 ft., and dew point decreases about 1°F/1,000 ft. In a convective current, temperature and dew point converge at about 4.4°F/1,000 ft. Thus, if the temperature and dew point are 70°F and 48°F, respectively, at 1,000 ft. MSL, there would be a 22°F spread that, divided by the lapse rate of 4.4, is approximately 5,000 ft. AGL, or 6,000 ft. MSL (5,000 + 1,000).

Answer (A) is incorrect. The approximate base of the cumulus clouds is 4,000 ft. MSL if the temperature at 1,000 ft. MSL is 61°F, not 70°F. **Answer (B) is incorrect.** The approximate base of the cumulus clouds is 5,000 ft. AGL, not MSL.

Answer (A) is correct. (AC 00-6B)

DISCUSSION: Cumulus clouds are formed in a convective updraft, build upward, and are associated with turbulence.

Answer (B) is incorrect. A temperature inversion prevents updrafts from forming, which is needed for the formation of cumulus clouds. **Answer (C) is incorrect.** The dry adiabatic lapse rate is a measurement of air with no moisture available to form clouds.

Answer (A) is correct. (AC 00-6B Chap 10)

DISCUSSION: Characteristics of a stable air mass include stratiform clouds, continuous precipitation, smooth air, and fair to poor visibility in haze and smoke.

Answer (B) is incorrect. Restricted, not unlimited, visibility is an indication of stable air. **Answer (C) is incorrect.** Fair weather cumulus clouds indicate unstable conditions, not stable conditions.

Answer (A) is correct. (AC 00-6B Chap 11)

DISCUSSION: Moist, stable air flowing upslope can be expected to produce stratus type clouds as the air cools adiabatically as it moves up sloping terrain.

Answer (B) is incorrect. Showers and thunderstorms are characteristics of unstable (not stable) air. **Answer (C) is incorrect.** Convective turbulence is a characteristic of unstable (not stable) air.

Answer (C) is correct. (AC 00-6B Chap 17)

DISCUSSION: When unstable air is lifted, it usually results in considerable vertical development and associated turbulence, i.e., convective activity.

Answer (A) is incorrect. Stable rather than unstable air creates stratus clouds with little vertical development. **Answer (B) is incorrect.** Stratus (layer-type) clouds usually have little turbulence unless they are either lenticular clouds created by mountain waves or other high-altitude clouds associated with high winds near or in the jet stream.

54. What are characteristics of unstable air?

- A. Turbulence and good surface visibility.
- B. Turbulence and poor surface visibility.
- C. Nimbostratus clouds and good surface visibility.

55. A stable air mass is most likely to have which characteristic?

- A. Showery precipitation.
- B. Turbulent air.
- C. Poor surface visibility.

56. Steady precipitation preceding a front is an indication of

- A. stratiform clouds with moderate turbulence.
- B. cumuliform clouds with little or no turbulence.
- C. stratiform clouds with little or no turbulence.

57. What are characteristics of a moist, unstable air mass?

- A. Cumuliform clouds and showery precipitation.
- B. Poor visibility and smooth air.
- C. Stratiform clouds and showery precipitation.

58. What measurement can be used to determine the stability of the atmosphere?

- A. Atmospheric pressure.
- B. Actual lapse rate.
- C. Surface temperature.

59. The stability of an air mass can usually be determined by

- A. the height of the tropopause.
- B. measuring the dry adiabatic lapse rate.
- C. cloud types and the type of precipitation.

Answer (A) is correct. (FAA-H-8083-25B Chap 12)

DISCUSSION: Characteristics of an unstable air mass include cumuliform clouds, showery precipitation, turbulence, and good visibility, except in blowing obstructions.

Answer (B) is incorrect. Poor surface visibility is a characteristic of stable (not unstable) air. **Answer (C) is incorrect.** Stratus clouds are characteristic of stable (not unstable) air.

Answer (C) is correct. (AC 00-6B Chap 12)

DISCUSSION: Characteristics of a stable air mass include stratiform clouds and fog, continuous precipitation, smooth air, and fair to poor visibility in haze and smoke.

Answer (A) is incorrect. Showery precipitation is a characteristic of an unstable (not stable) air mass. **Answer (B) is incorrect.** Turbulent air is a characteristic of an unstable (not stable) air mass.

Answer (C) is correct. (AC 00-6B Chap 10)

DISCUSSION: Steady precipitation preceding a front is usually an indication of a warm front, which results from warm air being cooled from the bottom by colder air. This results in stratiform clouds with little or no turbulence.

Answer (A) is incorrect. Stratiform clouds usually are not turbulent. **Answer (B) is incorrect.** Cumuliform clouds have showery rather than steady precipitation.

Answer (A) is correct. (FAA-H-8083-25B Chap 12)

DISCUSSION: Characteristics of an unstable air mass include cumuliform clouds, showery precipitation, turbulence, and good visibility, except in blowing obstructions.

Answer (B) is incorrect. Poor visibility and smooth air are characteristics of stable (not unstable) air. **Answer (C) is incorrect.** Stratiform clouds and continuous precipitation are characteristics of stable (not unstable) air.

Answer (B) is correct. (AC 00-6B Chap 12)

DISCUSSION: Atmospheric stability is determined by vertical movements of air. Warm air rises when the air above is cooler. The actual lapse rate, which is the decrease of temperature with altitude, is therefore a measure of stability.

Answer (A) is incorrect. Atmospheric pressure is the pressure exerted by the atmosphere as a consequence of gravitational attraction exerted upon the "column" of air lying directly above the point in question. It cannot be used to determine stability. **Answer (C) is incorrect.** While the surface temperature may have some effect on temperature changes and air movements, it is the actual lapse rate that determines the stability of the atmosphere.

Answer (C) is correct. (FAA-H-8083-25B Chap 12)

DISCUSSION: Atmospheric stability influences weather by affecting the vertical motion of air. Stable air suppresses vertical motion, but unstable air enhances it. Clouds formed in stable air will be shallow and layered, e.g., stratus clouds. Clouds formed in unstable air will have more height and be of the cumulus or cumulonimbus type. Precipitation from stratus clouds tends to be over large areas and lasts for long periods. Precipitation from cumulus clouds tends to be more intense and lasts for short periods.

Answer (A) is incorrect. The tropopause is a thin boundary area between the troposphere and the stratosphere. The height of the tropopause varies widely due to the location above the earth and the time of the year and is not necessarily indicative of air mass stability. **Answer (B) is incorrect.** Stability is determined by the change in the ambient lapse rate, not the dry lapse rate.

60. What would decrease the stability of an air mass?

- A. Warming from below.
- B. Cooling from below.
- C. Decrease in water vapor.

61. What are the characteristics of an unstable atmosphere?

- A. A cool, dry air mass.
- B. A warm, humid air mass.
- C. Descending air in the northern hemisphere.

62. Clouds with extensive vertical development over mountainous terrain are a sign of

- A. a dry adiabatic lapse rate.
- B. a stable air mass.
- C. an unstable air mass.

7.11 Temperature Inversions

63. What feature is associated with a temperature inversion?

- A. A stable layer of air.
- B. An unstable layer of air.
- C. Chinook winds on mountain slopes.

64. The most frequent type of ground or surface-based temperature inversion is that which is produced by

- A. terrestrial radiation on a clear, relatively still night.
- B. warm air being lifted rapidly aloft in the vicinity of mountainous terrain.
- C. the movement of colder air under warm air, or the movement of warm air over cold air.

Answer (A) is correct. (AC 00-6B Chap 12)

DISCUSSION: When air is warmed from below, even though cooling adiabatically, it remains warmer than the surrounding air. The colder, more dense surrounding air forces the warmer air upward, and an unstable condition develops.

Answer (B) is incorrect. Cooling from below means the surrounding air is warmer, which would increase (not decrease) the stability of an air mass. **Answer (C) is incorrect.** As water vapor in air decreases, the air mass tends to increase (not decrease) stability.

Answer (B) is correct. (FAA-H-8083-25B Chap 12)

DISCUSSION: The stability of the atmosphere depends on its ability to resist vertical motion. As air temperature and air moisture increase, the density of the air decreases, causing it to rise. This creates an unstable atmosphere in which small vertical air movements tend to become larger, resulting in turbulent airflow and convective activity.

Answer (A) is incorrect. When air is cool, it resists rising, resulting in stability. **Answer (C) is incorrect.** The characteristics of a stable atmosphere do not change whether you are in the northern or southern hemisphere.

Answer (C) is correct. (AC 00-6B)

DISCUSSION: Winds across mountains cause mountain waves that are associated with severe turbulence, strong vertical currents, and icing. The extent of the turbulence is relative to the height of the ground, speed of the wind, and instability of the atmosphere. With adequate moisture, lenticular clouds will form at the top of each wave.

Answer (A) is incorrect. The dry adiabatic lapse rate is a measurement of air with no moisture available to form clouds.

Answer (B) is incorrect. The vertical development of the clouds indicates the presence of multiple waves with adequate upward motion and moisture to cause cloud formation.

Answer (A) is correct. (AC 00-6B Chap 2)

DISCUSSION: A temperature inversion is associated with an increase in temperature with height, a reversal of normal decrease in temperature with height. Thus, any warm air rises to where it is the same temperature and forms a stable layer of air.

Answer (B) is incorrect. Instability is a result of rising air remaining warmer than the surrounding air aloft, which would not occur with a temperature inversion. **Answer (C) is incorrect.** A Chinook wind is a warm, dry downslope wind blowing down the eastern slopes of the Rocky Mountains over the adjacent plains in the U.S. and Canada.

Answer (A) is correct. (AC 00-6B Chap 16)

DISCUSSION: An inversion often develops near the ground on clear, cool nights when wind is light. The ground loses heat and cools the air near the ground while the temperature a few hundred feet above changes very little. Thus, temperature increases in height, which is an inversion.

Answer (B) is incorrect. Warm air being lifted rapidly aloft in the vicinity of mountainous terrain describes convective activity. **Answer (C) is incorrect.** The movement of colder air under warm air, which causes an inversion, is caused by a cold front, not terrestrial radiation (warm air moving over cold air is a warm front).

65. When there is a temperature inversion, you would expect to experience

- A. clouds with extensive vertical development above an inversion aloft.
- B. good visibility in the lower levels of the atmosphere and poor visibility above an inversion aloft.
- C. an increase in temperature as altitude increases.

66. Which weather conditions should be expected beneath a low-level temperature inversion layer when the relative humidity is high?

- A. Smooth air, poor visibility, fog, haze, or low clouds.
- B. Light wind shear, poor visibility, haze, and light rain.
- C. Turbulent air, poor visibility, fog, low stratus type clouds, and showery precipitation.

67. An increase in temperature with an altitude increase

- A. is indication of an inversion.
- B. denotes the beginning of the stratosphere.
- C. means a cold front passage.

68. Temperature and radiation variations over land with a clear sky typically lead to

- A. minimum temperature occurring after sunrise.
- B. outgoing terrestrial radiation peaking at noon.
- C. temperature reaching a maximum closer to noon than to sunset.

Answer (C) is correct. (AC 00-6B Chap 17)

DISCUSSION: By definition, a temperature inversion is a situation in which the temperature increases as altitude increases. The normal situation is that the temperature decreases as altitude increases.

Answer (A) is incorrect. Vertical development does not occur in an inversion situation because the warm air cannot rise when the air above is warmer. **Answer (B) is incorrect.** The inversion traps dust, smoke, and other nuclei beneath the inversion, which reduces visibility.

Answer (A) is correct. (AC 00-6B Chap 21)

DISCUSSION: Beneath temperature inversions, there is usually smooth air because there is little vertical movement due to the inversion. There is also poor visibility due to fog, haze, and low clouds (when there is high relative humidity).

Answer (B) is incorrect. Wind shears usually do not occur below a low-level temperature inversion. They occur at or just above the inversion. **Answer (C) is incorrect.** Turbulent air and showery precipitation are not present with low-level temperature inversions.

Answer (A) is correct. (FAA-H-8083-25B Chap 12)

DISCUSSION: Normally, as air rises and expands in the atmosphere, the temperature decreases. However, when the temperature of the air rises with altitude, this indicates that a temperature inversion exists.

Answer (B) is incorrect. Although the temperature does begin to increase in the stratosphere, only specialized aircraft are likely to be able to operate high enough to see this change.

Answer (C) is incorrect. A cold frontal passage does not always cause the temperature to increase with altitude. If it did, this would be an indication of an inversion.

Answer (A) is correct. (AC 00-6B)

DISCUSSION: At night, heating is absent, but terrestrial radiation continues cooling the earth's surface. Cooling continues until shortly after sunrise, when incoming solar radiation once again exceeds outgoing terrestrial radiation. Minimum surface air temperature usually occurs shortly after sunrise.

Answer (B) is incorrect. Outgoing radiation peaks during the nighttime when no isolation occurs. **Answer (C) is incorrect.** Peak isolation occurs around noon, but maximum surface air temperature occurs during mid-afternoon.

STUDY UNIT EIGHT
AVIATION WEATHER SERVICES

(9 pages of outline)

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This study unit contains outlines of major concepts tested, sample test questions and answers regarding aviation weather services, and an explanation of each answer. The table of contents above lists each subunit within this study unit, the number of questions pertaining to that particular subunit, and the pages on which the outlines and questions begin, respectively.



Recall that the **sole purpose** of this book is to expedite your passing of the FAA pilot knowledge test for the private pilot certificate. Accordingly, all extraneous material (i.e., topics or regulations not directly tested on the FAA pilot knowledge test) is omitted, even though much more knowledge is necessary to fly safely. This additional material is presented in *Pilot Handbook* and *Private Pilot Flight Maneuvers and Practical Test Prep*, available from Gleim Publications, Inc. Order online at www.GleimAviation.com.